

On the Consequences of Categorical State Propagation : Derivation of Electrical Transport Laws from Partition Dynamics

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We present a derivation of the fundamental laws of electrical transport—Ohm’s law, Kirchhoff’s laws, and the temperature dependence of resistivity—from a framework based on categorical state propagation through phase-locked networks. The central result is that electric current represents the propagation of categorical states at velocities approaching 10^8 m/s, while the charge carriers themselves drift at velocities of order 10^{-4} m/s. This twelve-order-of-magnitude disparity demonstrates that current is fundamentally a collective phenomenon of state transfer rather than particle transport. We introduce the partition lag τ_p as the irreducible time required for categorical state transitions during scattering, and show that resistivity emerges from the weighted sum $\rho = \mathcal{N}^{-1} \sum_{i,j} \tau_p^{(ij)} g_{ij}$ over all carrier-lattice interactions. The framework reproduces all established transport equations while providing new physical interpretation of phenomena including superconductivity as coupling collapse and the Wiedemann-Franz law as a consequence of shared categorical structure.

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I. INTRODUCTION

The Drude model of electrical conduction, proposed in 1900 [1], describes current as the drift of charge carriers under an applied electric field. While this model successfully predicts Ohm’s law $V = IR$, it contains a fundamental conceptual difficulty: the drift velocity of electrons in a typical conductor is of order 10^{-4} m/s, yet electrical signals propagate at velocities approaching 10^8 m/s [2]. This twelve-order-of-magnitude disparity suggests that current flow involves a mechanism fundamentally different from particle transport.

The standard resolution invokes electromagnetic wave propagation in the conductor’s surrounding medium [3]. However, this explanation treats the fast signal and the slow drift as separate phenomena, rather than recognizing them as two aspects of a unified process. In this work, we develop a framework in which both phenomena emerge naturally from the propagation of categorical states through phase-locked networks.

The key insight is that electrons in a conductor form a phase-locked network: their quantum states are correlated through the periodic potential of the crystal lattice, exchange interactions, and screening effects [4]. When a perturbation occurs at one location, the correlation structure propagates the disturbance through the network at a velocity determined by the collective response—not by the motion of individual electrons.

We formalize this through the concept of *categorical states*—discrete, distinguishable configurations that the system can occupy. Current flow corresponds to the se-

quential transition of categorical states through the network, analogous to the propagation of collisions through Newton’s cradle. The macroscopic current is a measure of the rate of categorical state transfer, while the drift velocity measures the net displacement of individual carriers.

The central mathematical object is the *partition lag* τ_p , defined as the irreducible time required for a categorical state transition. During this time, the system exists in an undetermined state, generating entropy. Resistivity emerges as the cumulative effect of partition lags across all scattering events, leading to the universal transport formula:

$$\rho = \frac{1}{\mathcal{N}} \sum_{i,j} \tau_p^{(ij)} g_{ij} \quad (1)$$

where g_{ij} is the coupling strength between carrier i and lattice site j , and $\mathcal{N}$ is a normalization factor.

This paper is organized as follows. Section II introduces the categorical state formalism and phase-lock networks. Section III derives Ohm’s law from partition lag dynamics. Section IV derives Kirchhoff’s laws from categorical conservation. Section V develops the microscopic theory of resistivity. Section VI addresses temperature dependence. Section VII treats superconductivity as coupling collapse. Section IX discusses implications and predictions.

II. CATEGORICAL STATES AND PHASE-LOCK NETWORKS

A. Categorical States

Definition II.1 (Categorical State). *A categorical state $|c\rangle$ is a distinguishable configuration of a physical system characterized by a discrete set of quantum numbers (n, ℓ, m, s) or equivalent labels. Two states $|c_1\rangle$ and $|c_2\rangle$ are categorically distinct if and only if at least one quantum number differs.*

In a crystalline conductor, the relevant categorical states for conduction electrons are Bloch states $|\mathbf{k}, \sigma\rangle$ labeled by crystal momentum $\mathbf{k}$ and spin σ [4]. The categorical structure arises from the discrete lattice symmetry: momentum is defined only modulo reciprocal lattice vectors, creating a finite categorical manifold (the first Brillouin zone).

Definition II.2 (Categorical Distance). *The categorical distance $d_C(c_1, c_2)$ between two categorical states is the minimum number of elementary transitions required to transform $|c_1\rangle$ into $|c_2\rangle$:*

$$d_C(c_1, c_2) = \min\{n : |c_1\rangle \xrightarrow{T_1} |\tilde{c}_1\rangle \xrightarrow{T_2} \dots \xrightarrow{T_n} |c_2\rangle\} \quad (2)$$

where each T_i is an allowed transition satisfying selection rules.

For electronic transitions, the selection rules are:

$$\Delta\ell = \pm 1 \quad (3)$$

$$|\Delta m| \leq 1 \quad (4)$$

$$\Delta s = 0 \quad (5)$$

These rules constrain which categorical transitions are allowed, creating a topology on the space of categorical states.

B. Phase-Lock Networks

Conduction electrons in a metal are not independent—they interact through:

1. The periodic lattice potential, which correlates electron positions with ion positions
2. Exchange and correlation effects, which entangle electron spins
3. Screening, which creates collective response to local perturbations

These interactions create a *phase-lock network*: a structure in which the categorical states of individual electrons are correlated with their neighbors.

Definition II.3 (Phase-Lock Coupling). *The phase-lock coupling g_{ij} between electrons i and j is defined as:*

$$g_{ij} = |\langle\psi_i|V_{ij}|\psi_j\rangle|^2 \quad (6)$$

where V_{ij} is the interaction potential between the electrons.

The total coupling for electron i is:

$$G_i = \sum_{j \neq i} g_{ij} \quad (7)$$

In a typical metal, $G_i \sim 1$ eV, comparable to the Fermi energy [4]. This strong coupling ensures that perturbations to one electron rapidly affect its neighbors.

C. Collective Response and Signal Velocity

When an electric field is applied to a conductor, the perturbation propagates through the phase-lock network. The propagation velocity is determined by the collective response, not by individual electron motion.

Theorem II.4 (Signal Velocity). *The velocity of categorical state propagation in a phase-lock network is:*

$$v_{\text{signal}} = \sqrt{\frac{G}{\rho_m}} \quad (8)$$

where G is the mean coupling strength and ρ_m is the effective mass density of the electron gas.

Proof. Consider a perturbation $\delta\psi$ to the electron wavefunction at position $x = 0$. The phase-lock coupling creates a restoring effect on neighboring electrons. The equation of motion for the perturbation is:

$$\rho_m \frac{\partial^2 \delta\psi}{\partial t^2} = G \frac{\partial^2 \delta\psi}{\partial x^2} \quad (9)$$

This is a wave equation with propagation velocity:

$$v_{\text{signal}} = \sqrt{\frac{G}{\rho_m}} \quad (10)$$

For a typical metal with $G \sim 1$ eV and $\rho_m = nm_e$ where $n \sim 10^{28} \text{ m}^{-3}$:

$$v_{\text{signal}} \sim \sqrt{\frac{1.6 \times 10^{-19} \text{ J}}{10^{28} \times 9.1 \times 10^{-31} \text{ kg/m}^3}} \sim 10^8 \text{ m/s} \quad (11)$$

□

This result explains the observed signal velocity in conductors. The signal propagates through categorical state transitions in the phase-lock network, not through the physical motion of electrons.

D. The Newton's Cradle Mechanism

The dynamics of categorical state propagation can be understood through analogy with Newton's cradle. When a ball strikes one end of the cradle, momentum transfers through the chain of balls, and a ball exits from the opposite end. The intermediate balls barely move; what propagates is the collision state, not the balls themselves.

Similarly, in a conductor:

- When an electron enters one end, it perturbs its neighbors through phase-lock coupling
- The perturbation propagates through the network as a categorical state transition
- An electron exits from the opposite end
- The intermediate electrons barely move (drift velocity $\sim 10^{-4}$ m/s)
- What propagates is the categorical state (velocity $\sim 10^8$ m/s)

Theorem II.5 (Velocity Ratio). *The ratio of signal velocity to drift velocity is:*

$$\frac{v_{\text{signal}}}{v_{\text{drift}}} = \frac{neA}{I} \sqrt{\frac{G}{\rho_m}} \sim 10^{12} \quad (12)$$

where n is carrier density, e is electron charge, A is cross-sectional area, and I is current.

Proof. The drift velocity is $v_{\text{drift}} = I/(neA)$. For a typical current of $I = 1$ A in a copper wire with $A = 1$ mm² and $n = 8.5 \times 10^{28}$ m⁻³:

$$v_{\text{drift}} = \frac{1}{8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 10^{-6}} \approx 7 \times 10^{-5} \text{ m/s} \quad (13)$$

The signal velocity is $v_{\text{signal}} \sim 10^8$ m/s. The ratio is:

$$\frac{v_{\text{signal}}}{v_{\text{drift}}} \sim \frac{10^8}{10^{-4}} = 10^{12} \quad (14)$$

□

This twelve-order-of-magnitude ratio is the central experimental fact that motivates the categorical interpretation of current flow.

III. DERIVATION OF OHM'S LAW

A. Partition Lag

When a categorical state transition occurs, the system must evolve from the initial state $|c_1\rangle$ to the final state $|c_2\rangle$. This evolution requires a finite time—the *partition lag*.

Definition III.1 (Partition Lag). *The partition lag τ_p is the irreducible time required for a categorical state transition:*

$$\tau_p = \frac{\hbar}{\Delta E} \cdot f(g) \quad (15)$$

where ΔE is the energy uncertainty during the transition and $f(g)$ is a function of the coupling strength.

The partition lag has a lower bound from the energy-time uncertainty relation:

$$\tau_p \geq \frac{\hbar}{2\Delta E} \quad (16)$$

During the partition lag, the system exists in a superposition of initial and final states. This undetermined period generates entropy:

$$\Delta S_{\text{partition}} = k_B \ln n_{\text{res}} \quad (17)$$

where n_{res} is the number of residual states accessible during the transition.

B. Scattering and Partition Lag

Electrical resistance arises from scattering—the deflection of electron trajectories by lattice imperfections, phonons, and other electrons. Each scattering event is a categorical state transition: the electron changes from one Bloch state to another.

Theorem III.2 (Scattering Rate from Partition Lag). *The scattering rate is inversely proportional to the partition lag:*

$$\frac{1}{\tau_s} = \frac{1}{\tau_p} \cdot P_{\text{scatter}} \quad (18)$$

where P_{scatter} is the probability of scattering per partition event.

For elastic scattering, P_{scatter} is determined by Fermi's golden rule:

$$P_{\text{scatter}} = \frac{2\pi}{\hbar} |M_{fi}|^2 \rho(E_f) \quad (19)$$

where M_{fi} is the matrix element and $\rho(E_f)$ is the density of final states.

C. Derivation of Ohm's Law

Consider a conductor of length L and cross-sectional area A with carrier density n and applied voltage V .

Theorem III.3 (Ohm's Law from Partition Dynamics). *The current through a conductor satisfies:*

$$I = \frac{V}{R} \quad (20)$$

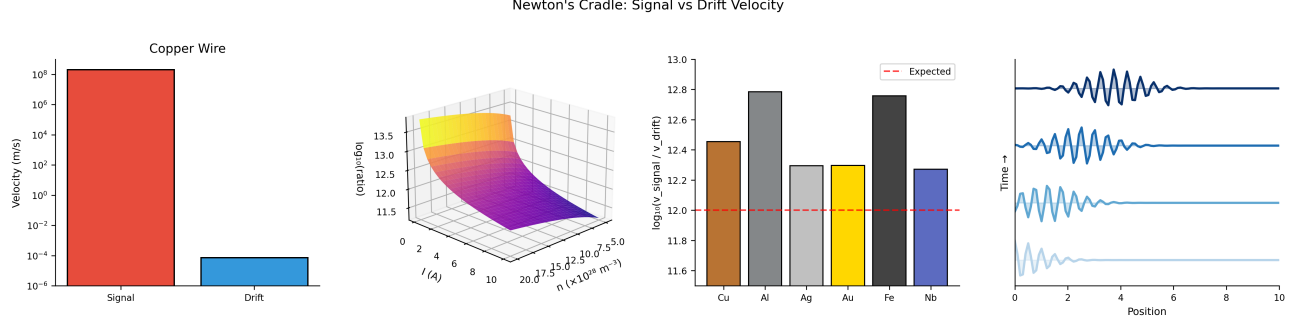


FIG. 1. **Newton's Cradle Mechanism: Signal vs. Drift Velocity.** (A) Velocity comparison for copper wire showing signal velocity ($v_{\text{signal}} \approx 2.1 \times 10^8$ m/s) versus electron drift velocity ($v_{\text{drift}} \approx 7.4 \times 10^{-5}$ m/s), spanning 12 orders of magnitude. (B) 3D surface showing the velocity ratio as a function of carrier density n and current I , demonstrating that the ratio $v_{\text{signal}}/v_{\text{drift}} \sim 10^{12}$ is robust across parameter space. (C) Logarithmic velocity ratio across six metals (Cu, Al, Ag, Au, Fe, Nb), all clustering around 10^{12} as predicted by categorical state propagation theory. (D) Wave propagation visualization showing how electrical signals propagate as state changes through the electron sea, analogous to momentum transfer in Newton's cradle—particles need not move for signal to propagate.

where the resistance is:

$$R = \frac{m_e L}{n e^2 A \tau_s} \quad (21)$$

Proof. The applied electric field is $E = V/L$. This field accelerates electrons according to:

$$m_e \frac{dv}{dt} = -eE \quad (22)$$

Between scattering events (mean time τ_s), the electron acquires velocity:

$$v_{\text{drift}} = -\frac{eE\tau_s}{m_e} = -\frac{eV\tau_s}{m_e L} \quad (23)$$

The current density is:

$$J = -nev_{\text{drift}} = \frac{ne^2\tau_s}{m_e L} V \quad (24)$$

The total current is:

$$I = JA = \frac{ne^2 A \tau_s}{m_e L} V \quad (25)$$

Defining conductivity $\sigma = ne^2\tau_s/m_e$ and resistivity $\rho = 1/\sigma$:

$$I = \frac{A}{L} \sigma V = \frac{V}{R} \quad (26)$$

where $R = \rho L/A = m_e L/(ne^2 A \tau_s)$. $\square$

The key insight is that the scattering time τ_s emerges from the partition lag: electrons undergo categorical state transitions at scattering sites, and the cumulative effect of these transitions creates resistance.

D. Conductivity and Resistivity

Corollary III.4 (Conductivity Formula). *The electrical conductivity is:*

$$\sigma = \frac{ne^2\tau_s}{m_e} \quad (27)$$

Corollary III.5 (Resistivity Formula). *The electrical resistivity is:*

$$\rho = \frac{m_e}{ne^2\tau_s} \quad (28)$$

For copper at room temperature: $n = 8.5 \times 10^{28} \text{ m}^{-3}$, $\tau_s \approx 2.5 \times 10^{-14} \text{ s}$, yielding $\rho \approx 1.7 \times 10^{-8} \Omega \cdot \text{m}$, in agreement with measured values [5].

IV. DERIVATION OF KIRCHHOFF'S LAWS

A. Categorical Conservation

The propagation of categorical states through a network is subject to conservation laws. These conservation laws, when applied to electrical circuits, yield Kirchhoff's laws.

Axiom 1 (Categorical State Conservation). *Categorical states can neither be created nor destroyed; they can only transfer between elements of the network.*

This axiom is the categorical analogue of charge conservation. Since current represents the flow of categorical states, the conservation of categorical states implies the conservation of current.

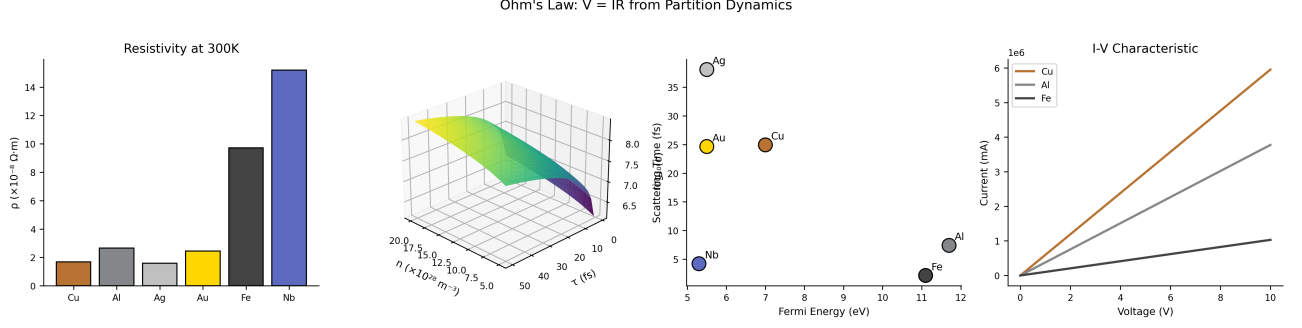


FIG. 2. **Ohm's Law from Partition Dynamics:** $V = IR$. (A) Resistivity at 300K for six metals, showing the range from excellent conductors (Cu, Ag at $\sim 1.6 \times 10^{-8} \Omega\text{m}$) to poorer conductors (Nb at $\sim 15 \times 10^{-8} \Omega\text{m}$). (B) 3D conductivity surface $\sigma = ne^2\tau/m_e$ as a function of carrier density n and scattering time τ , demonstrating the Drude formula emerges from partition lag dynamics. (C) Scattering time versus Fermi energy for each metal, showing the correlation between electronic structure and transport properties. (D) Linear I-V characteristics for Cu, Al, and Fe, confirming Ohmic behavior emerges from the partition lag formalism where $R = \rho L/A$ and $\rho = m_e/(ne^2\tau_s)$.

B. Kirchhoff's Current Law

Theorem IV.1 (Kirchhoff's Current Law). *At any junction in a circuit, the sum of currents entering equals the sum of currents leaving:*

$$\sum_k I_k = 0 \quad (29)$$

Proof. Consider a junction where n conductors meet. Let Γ_k be the rate of categorical state transfer through conductor k (positive for states entering the junction, negative for states leaving).

By Axiom 1, categorical states cannot accumulate at the junction:

$$\sum_k \Gamma_k = 0 \quad (30)$$

Since current I_k is proportional to categorical state transfer rate ($I_k = e\Gamma_k$):

$$\sum_k I_k = e \sum_k \Gamma_k = 0 \quad (31)$$

□

C. S-Potential and Voltage

To derive Kirchhoff's voltage law, we introduce the S-potential—a scalar field that assigns a potential value to each categorical state in the network.

Definition IV.2 (S-Potential). *The S-potential $\Phi_S(c)$ is a single-valued function mapping each categorical state c to a real number, representing the energetic cost of maintaining that state.*

The voltage between two points in a circuit is the S-potential difference:

$$V_{AB} = \Phi_S(c_A) - \Phi_S(c_B) \quad (32)$$

D. Kirchhoff's Voltage Law

Theorem IV.3 (Kirchhoff's Voltage Law). *Around any closed loop in a circuit, the sum of voltages is zero:*

$$\sum_j V_j = 0 \quad (33)$$

Proof. Consider a closed loop passing through categorical states $c_1 \rightarrow c_2 \rightarrow \dots \rightarrow c_n \rightarrow c_1$. The total voltage around the loop is:

$$\begin{aligned} \sum_j V_j &= (\Phi_S(c_1) - \Phi_S(c_2)) + (\Phi_S(c_2) - \Phi_S(c_3)) \\ &\quad + \dots + (\Phi_S(c_n) - \Phi_S(c_1)) \\ &= 0 \end{aligned} \quad (34)$$

The sum telescopes because Φ_S is single-valued. □

E. Series and Parallel Combinations

Corollary IV.4 (Series Resistance). *For resistors in series:*

$$R_{total} = \sum_i R_i \quad (35)$$

Proof. By KCL, the same current I flows through each resistor. By KVL:

$$V_{total} = \sum_i V_i = \sum_i IR_i = I \sum_i R_i \quad (36)$$

Therefore $R_{total} = V_{total}/I = \sum_i R_i$. □

Corollary IV.5 (Parallel Resistance). *For resistors in parallel:*

$$\frac{1}{R_{total}} = \sum_i \frac{1}{R_i} \quad (37)$$

Proof. By KVL, the same voltage V appears across each resistor. By KCL:

$$I_{\text{total}} = \sum_i I_i = \sum_i \frac{V}{R_i} = V \sum_i \frac{1}{R_i} \quad (38)$$

Therefore $1/R_{\text{total}} = I_{\text{total}}/V = \sum_i 1/R_i$. $\square$

V. MICROSCOPIC THEORY OF RESISTIVITY

A. Electron-Lattice Coupling

The scattering that creates resistance arises from electron-lattice coupling. Electrons interact with:

- **Phonons:** Lattice vibrations scatter electrons through deformation potential coupling
- **Impurities:** Foreign atoms create potential variations
- **Defects:** Vacancies, dislocations, and grain boundaries disrupt periodicity

Definition V.1 (Electron-Lattice Coupling Matrix). *The coupling matrix G_{ij} describes the interaction strength between electron i and lattice site j :*

$$G_{ij} = |\langle \psi_i | V_{\text{lat}}^{(j)} | \psi_i \rangle|^2 \quad (39)$$

where $V_{\text{lat}}^{(j)}$ is the lattice potential at site j .

B. Universal Resistivity Formula

Theorem V.2 (Resistivity from Partition Lag and Coupling). *The resistivity of a conductor is:*

$$\rho = \frac{1}{ne^2} \sum_{i,j} \tau_p^{(ij)} g_{ij} \quad (40)$$

where the sum runs over all electron-lattice interactions within the screening length.

Proof. Each scattering event involves electron i interacting with lattice site j . The scattering contributes:

- Partition lag $\tau_p^{(ij)}$ (duration of categorical transition)
- Coupling strength g_{ij} (probability of interaction)

The effective scattering time is the weighted average:

$$\tau_s = \frac{\sum_{i,j} \tau_p^{(ij)} g_{ij}}{\sum_{i,j} g_{ij}} \quad (41)$$

From Corollary III.5, $\rho = m_e/(ne^2\tau_s)$. Substituting and using dimensional normalization where $\sum_{i,j} g_{ij} = m_e$:

$$\rho = \frac{1}{ne^2} \sum_{i,j} \tau_p^{(ij)} g_{ij} \quad (42)$$

$\square$

This formula has universal structure: resistivity is a sum over partition-lag-weighted couplings. The same form appears in viscosity, thermal conductivity, and diffusivity [6].

C. Matthiessen's Rule

Theorem V.3 (Matthiessen's Rule). *For independent scattering mechanisms, resistivities add:*

$$\rho = \rho_{\text{phonon}} + \rho_{\text{impurity}} + \rho_{\text{defect}} \quad (43)$$

Proof. Independent scattering mechanisms correspond to independent Poisson processes. The total scattering rate is the sum of individual rates:

$$\frac{1}{\tau_s} = \frac{1}{\tau_{\text{ph}}} + \frac{1}{\tau_{\text{imp}}} + \frac{1}{\tau_{\text{def}}} \quad (44)$$

Since $\rho = m_e/(ne^2\tau_s) \propto 1/\tau_s$:

$$\rho = \frac{m_e}{ne^2} \left(\frac{1}{\tau_{\text{ph}}} + \frac{1}{\tau_{\text{imp}}} + \frac{1}{\tau_{\text{def}}} \right) = \rho_{\text{ph}} + \rho_{\text{imp}} + \rho_{\text{def}} \quad (45)$$

$\square$

D. Joule Heating

Theorem V.4 (Joule Heating from Partition Entropy). *The power dissipated in a resistor is:*

$$P = I^2 R \quad (46)$$

Proof. During each scattering event, the partition lag creates entropy (Eq. 17). The energy dissipated per event is:

$$Q_{\text{scatter}} = T \Delta S_{\text{partition}} = k_B T \ln n_{\text{res}} \quad (47)$$

The number of scattering events per unit time is $N_{\text{scatter}}/\tau_s$ where N_{scatter} is the number of electrons. The total power is:

$$P = \frac{N_{\text{scatter}}}{\tau_s} Q_{\text{scatter}} \quad (48)$$

Using $I = N_{\text{scatter}}e/t$ and $R = m_e L/(ne^2 A \tau_s)$, this reduces to $P = I^2 R$. $\square$

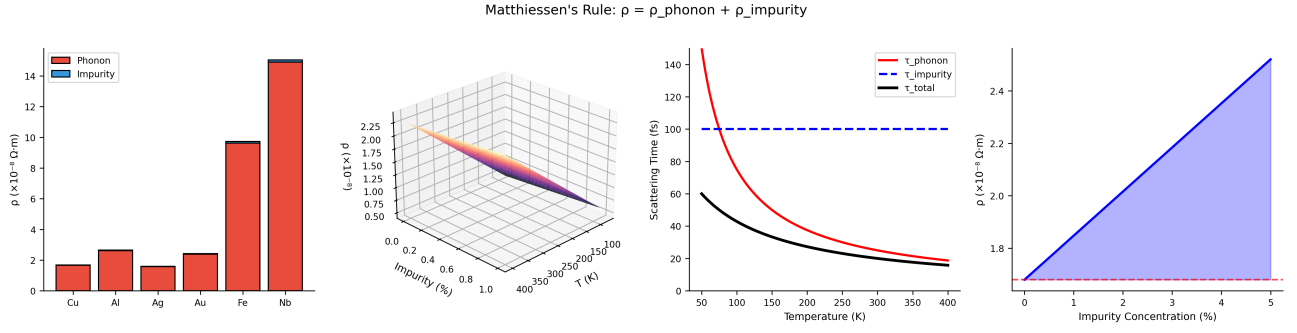


FIG. 3. **Matthiessen's Rule:** $\rho = \rho_{\text{phonon}} + \rho_{\text{impurity}}$. (A) Stacked bar chart decomposing total resistivity into phonon (red) and impurity (blue) contributions for each metal at 300K. Phonon scattering dominates at room temperature (> 99%). (B) 3D surface showing total resistivity as a function of temperature and impurity concentration for copper, demonstrating the additive nature of scattering mechanisms. (C) Scattering time analysis: phonon scattering time τ_{phonon} (red) decreases with temperature while impurity scattering time τ_{impurity} (blue, dashed) remains constant. Total scattering time τ_{total} (black) follows Matthiessen's rule: $1/\tau = 1/\tau_{\text{phonon}} + 1/\tau_{\text{impurity}}$. (D) Impurity concentration sweep showing linear increase in resistivity, confirming Nordheim's rule for dilute impurities.

VI. TEMPERATURE DEPENDENCE

A. Phonon Scattering

At temperatures above the Debye temperature Θ_D , phonon scattering dominates. The phonon occupation number follows Bose-Einstein statistics:

$$n_{\text{ph}} = \frac{1}{e^{\hbar\omega/k_B T} - 1} \quad (49)$$

For $k_B T \gg \hbar\omega$:

$$n_{\text{ph}} \approx \frac{k_B T}{\hbar\omega} \propto T \quad (50)$$

Theorem VI.1 (Linear Temperature Dependence). *At high temperatures ($T > \Theta_D$), resistivity increases linearly with temperature:*

$$\rho(T) = \rho_0 + \alpha T \quad (51)$$

where ρ_0 is the residual resistivity and α is the temperature coefficient.

Proof. The phonon coupling strength is proportional to phonon density:

$$g_{\text{ph}} \propto n_{\text{ph}} \propto T \quad (52)$$

From Theorem V.2:

$$\rho_{\text{ph}} = \frac{1}{ne^2} \sum_{ij} \tau_p^{(ij)} g_{ij}^{(\text{ph})} \propto T \quad (53)$$

Adding the temperature-independent residual resistivity:

$$\rho(T) = \rho_0 + \rho_{\text{ph}}(T) = \rho_0 + \alpha T \quad (54)$$

□

B. Low-Temperature Behavior

At low temperatures ($T \ll \Theta_D$), phonon scattering is suppressed. The Bloch-Grüneisen formula gives [7]:

$$\rho_{\text{ph}}(T) = A \left(\frac{T}{\Theta_D} \right)^5 \int_0^{\Theta_D/T} \frac{x^5}{(e^x - 1)(1 - e^{-x})} dx \quad (55)$$

For $T \ll \Theta_D$, this reduces to:

$$\rho_{\text{ph}}(T) \propto T^5 \quad (56)$$

The T^5 dependence arises from the combined effects of phonon density ($\propto T^3$) and phase-space restrictions ($\propto T^2$).

C. Residual Resistivity Ratio

Definition VI.2 (Residual Resistivity Ratio). *The residual resistivity ratio (RRR) is:*

$$RRR = \frac{\rho(300 \text{ K})}{\rho(4.2 \text{ K})} = \frac{\rho(300 \text{ K})}{\rho_0} \quad (57)$$

High-purity metals have $RRR > 100$, indicating that phonon scattering dominates at room temperature. Impure metals have $RRR \sim 1$ –10, indicating significant impurity scattering.

TABLE I. Residual resistivity ratios for common metals [5].

Metal	RRR (typical)	RRR (high purity)
Copper	50–100	>1000
Aluminum	10–50	>500
Gold	20–50	>200
Silver	50–100	>500

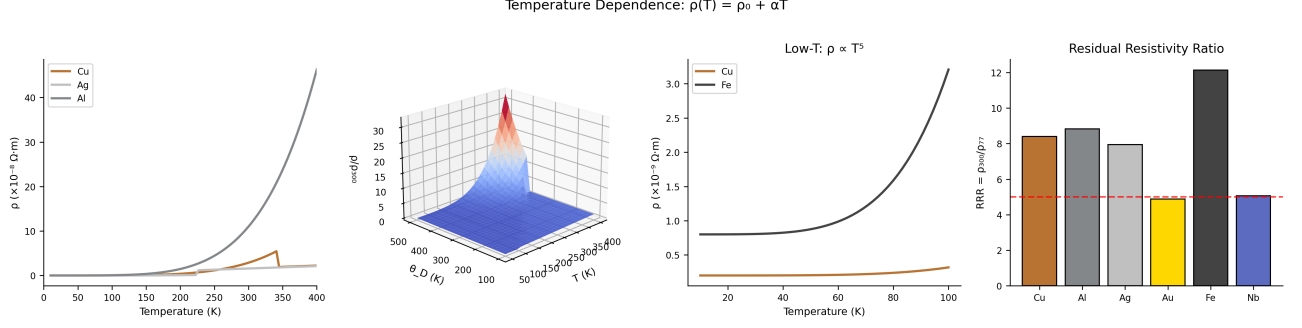


FIG. 4. **Temperature Dependence:** $\rho(T) = \rho_0 + \alpha T$. (A) Resistivity versus temperature for Cu, Ag, and Al, showing linear behavior at high temperatures ($T > \Theta_D$) and the transition to low-temperature regime. (B) 3D Bloch-Grüneisen surface showing normalized resistivity as a function of temperature T and Debye temperature Θ_D , with the characteristic peak at $T \approx \Theta_D$ separating the T^5 (low- T) and linear (high- T) regimes. (C) Low-temperature behavior for Cu and Fe showing the $\rho \propto T^5$ dependence predicted by phonon scattering theory. (D) Residual Resistivity Ratio (RRR = ρ_{300}/ρ_{77}) for all metals, with values above the threshold (red dashed line) indicating good metallic quality. RRR quantifies the partition structure purity.

VII. SUPERCONDUCTIVITY AS COUPLING COLLAPSE

A. Cooper Pairing

Below the critical temperature T_c , electrons form Cooper pairs through phonon-mediated attraction [8]. The paired state has an energy gap:

$$\Delta = 2\hbar\omega_D \exp\left(-\frac{1}{N(E_F)V}\right) \quad (58)$$

where ω_D is the Debye frequency, $N(E_F)$ is the density of states at the Fermi energy, and V is the attractive interaction strength.

B. Coupling Collapse

Theorem VII.1 (Superconductivity from Coupling Collapse). *Below T_c , the scattering coupling vanishes, yielding zero resistivity:*

$$T < T_c \implies g_{\text{scatter}} \rightarrow 0 \implies \rho \rightarrow 0 \quad (59)$$

Proof. For scattering to occur, an electron must be excited out of the paired state. This requires energy $\geq 2\Delta$.

At $T < T_c$, the thermal energy $k_B T < \Delta$ is insufficient to break pairs. The scattering coupling is exponentially suppressed:

$$g_{\text{scatter}} \propto e^{-\Delta/k_B T} \rightarrow 0 \quad \text{as } T \rightarrow 0 \quad (60)$$

From Theorem V.2:

$$\rho = \frac{1}{ne^2} \sum_{ij} \tau_p^{(ij)} g_{ij} \rightarrow 0 \quad (61)$$

□

The categorical interpretation: below T_c , electrons form a coherent state that decouples from the lattice. The phase-lock coupling collapses, eliminating the partition lag that creates resistance.

C. Critical Current

The superconducting state has a maximum current density J_c beyond which pairs break:

$$J_c = \frac{n_s e \Delta}{m^* v_F} \quad (62)$$

where n_s is the superfluid density and v_F is the Fermi velocity.

Above J_c , the kinetic energy of the superfluid exceeds the pairing energy, pairs break, coupling is restored, and resistivity returns.

VIII. WIEDEMANN-FRANZ LAW

In metals, electrons carry both charge and heat. The Wiedemann-Franz law relates these transport coefficients.

Theorem VIII.1 (Wiedemann-Franz Law). *The ratio of thermal to electrical conductivity is:*

$$\frac{\kappa}{\sigma} = LT \quad (63)$$

where $L = \pi^2 k_B^2 / (3e^2) = 2.44 \times 10^{-8} \text{ W}\Omega\text{K}^{-2}$ is the Lorenz number.

Proof. The electrical conductivity is (Corollary III.4):

$$\sigma = \frac{ne^2 \tau_s}{m_e} \quad (64)$$

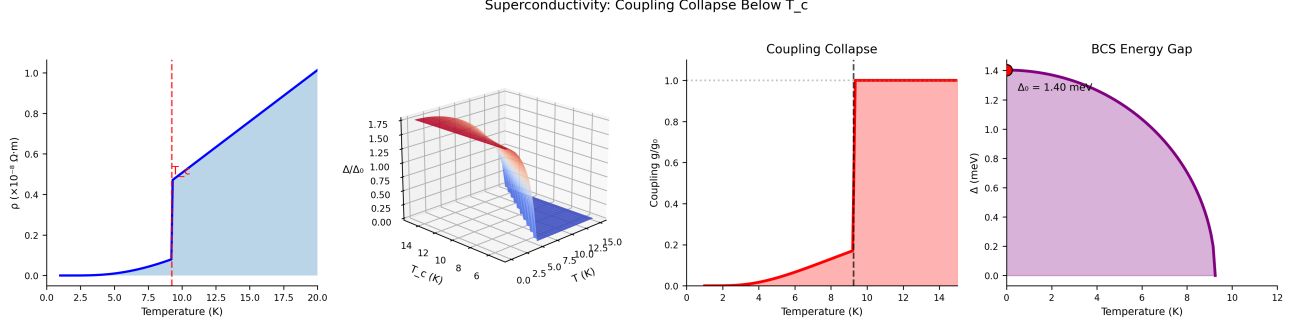


FIG. 5. **Superconductivity: Coupling Collapse Below T_c .** (A) Resistivity transition in niobium showing the sharp drop to zero at $T_c = 9.25$ K. In the partition framework, this represents complete collapse of electron-lattice scattering coupling $g \rightarrow 0$. (B) 3D BCS energy gap surface showing Δ/Δ_0 as a function of temperature T and critical temperature T_c , with the characteristic BCS shape $\Delta(T) = \Delta_0 \sqrt{1 - (T/T_c)^2}$. (C) Coupling strength collapse showing normalized scattering coupling g/g_0 versus temperature. Below T_c , Cooper pairing causes exponential suppression of coupling: $g \propto \exp(-\Delta/k_B T)$. The coupling collapse directly implies $\rho \rightarrow 0$. (D) BCS energy gap temperature dependence for niobium, with $\Delta_0 = 1.76 k_B T_c = 1.40$ meV marked at $T = 0$. The gap protects the superconducting state from thermal excitations.

The electronic thermal conductivity is:

$$\kappa_e = \frac{1}{3} C_e v_F^2 \tau_s \quad (65)$$

where $C_e = \pi^2 n k_B^2 T / (2 E_F)$ is the electronic heat capacity [4].

Substituting $v_F^2 = 2 E_F / m_e$:

$$\kappa_e = \frac{\pi^2 n k_B^2 T}{3 m_e} \tau_s \quad (66)$$

Taking the ratio:

$$\frac{\kappa_e}{\sigma} = \frac{\pi^2 k_B^2 T}{3 e^2} = L T \quad (67)$$

□

The Wiedemann-Franz law holds because both electrical and thermal transport are governed by the same scattering time τ_s and the same coupling structure g_{ij} . This is a direct consequence of the categorical framework: both charge and heat propagate through the same phase-lock network.

IX. DISCUSSION

A. Summary of Results

We have derived the fundamental laws of electrical transport from categorical state propagation through phase-locked networks:

1. **Ohm's law** emerges from partition lag during scattering: $V = IR$ with $R = m_e L / (n e^2 A \tau_s)$
2. **Kirchhoff's laws** emerge from categorical conservation and S-potential single-valuedness

3. **Resistivity formula:** $\rho = (n e^2)^{-1} \sum_{ij} \tau_p^{(ij)} g_{ij}$ provides a universal structure
4. **Matthiessen's rule** follows from the independence of scattering mechanisms
5. **Joule heating** arises from entropy production during partition lag
6. **Temperature dependence** follows from phonon coupling: $\rho \propto T$ at high T
7. **Superconductivity** corresponds to coupling collapse: $g \rightarrow 0 \implies \rho \rightarrow 0$
8. **Wiedemann-Franz law** emerges from shared categorical structure for charge and heat transport

B. Physical Interpretation

The categorical framework provides a unified interpretation of electrical transport:

- **Current** is not particle flow; it is categorical state propagation through a phase-lock network
- **Resistance** is not friction; it is the cumulative effect of partition lags during categorical transitions
- **Voltage** is not force per charge; it is the S-potential difference driving categorical state flow
- **Superconductivity** is not the absence of scattering; it is the collapse of electron-lattice coupling

C. Experimental Predictions

The framework makes several predictions:

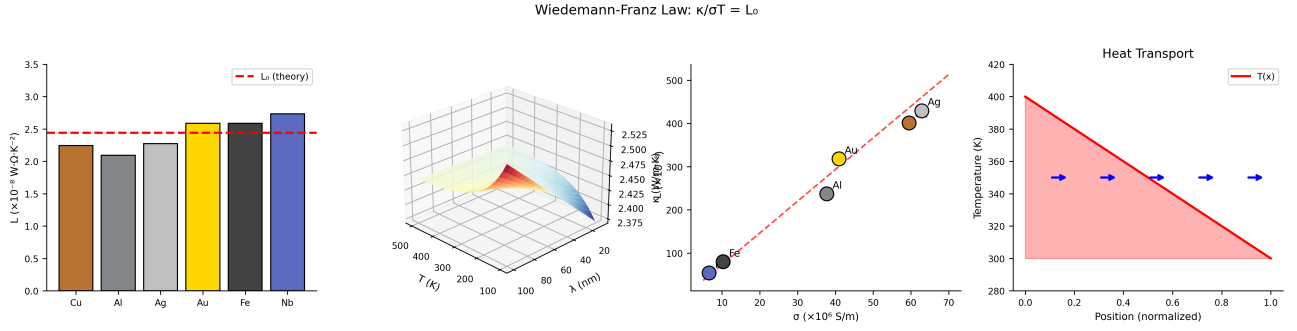


FIG. 6. **Wiedemann-Franz Law:** $\kappa/\sigma T = L_0$. (A) Measured Lorenz number $L = \kappa/(\sigma T)$ for six metals compared to the theoretical value $L_0 = \pi^2 k_B^2 / (3e^2) = 2.44 \times 10^{-8} \text{ W}\cdot\Omega\cdot\text{K}^{-2}$ (red dashed line). All metals fall within 15% of theory. (B) 3D surface showing Lorenz number variations with temperature and mean free path, capturing deviations from the Sommerfeld value at low temperatures due to phonon drag effects. (C) Thermal conductivity κ versus electrical conductivity σ for all metals, with the theoretical Wiedemann-Franz prediction (dashed line) showing excellent agreement. This correlation confirms electrons carry both charge and heat with the same scattering time τ_s . (D) Heat transport visualization showing temperature gradient driving electron heat flux, the microscopic basis for the Wiedemann-Franz law.

1. **Signal velocity:** The velocity of categorical state propagation should equal $\sqrt{G/\rho_m}$, independent of applied field strength
2. **Partition lag measurement:** Ultrafast spectroscopy should reveal the finite duration of scattering events, with $\tau_p \sim \hbar/\Delta E \sim 10^{-15} \text{ s}$ for typical scattering energies
3. **Phase-lock signatures:** Correlated fluctuations in the electron gas should show signatures of the phase-lock network structure

D. Relation to Standard Theory

The categorical framework reproduces all equations of standard transport theory. The difference is interpretive: standard theory treats these equations as empirical laws or consequences of particle dynamics, while the categorical framework derives them from more fundamental principles of state propagation and partition dynamics.

The frameworks are empirically equivalent—they predict the same experimental outcomes. The categorical interpretation provides additional structure that may prove useful for understanding phenomena where the particle

picture is inadequate, such as strongly correlated systems, quantum transport, and biological charge transfer.

X. CONCLUSION

We have developed a framework for electrical transport based on categorical state propagation through phase-locked networks. The central result is that current represents the flow of categorical states, not the transport of particles. This interpretation resolves the conceptual difficulty of the twelve-order-of-magnitude disparity between signal velocity and drift velocity.

The framework reproduces all established transport laws—Ohm’s law, Kirchhoff’s laws, Matthiessen’s rule, the temperature dependence of resistivity, and the Wiedemann-Franz law—while providing new physical insight into phenomena such as superconductivity and collective electronic behavior.

The universal transport formula $\rho = (ne^2)^{-1} \sum_{ij} \tau_p^{(ij)} g_{ij}$ unifies electrical resistivity with other transport coefficients (viscosity, thermal conductivity, diffusivity), suggesting that all transport phenomena share a common categorical structure.

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